

STATISTICAL ANALYSIS REPORT

Customer Sales Analysis Project

1. Introduction

The Customer Sales Analysis Project was developed to analyze customer purchasing behavior, identify top customers, evaluate sales performance, and generate business insights using Python and Pandas.

The project focuses on:

- Customer purchasing patterns
 - Sales trend analysis
 - Regional performance analysis
 - Customer retention analysis
 - Product performance analysis
 - Dashboard visualization
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2. Project Objectives

The main objectives of this project are:

- Analyze customer purchasing behavior
 - Identify top revenue-generating customers
 - Find best-selling products
 - Evaluate monthly and regional sales trends
 - Analyze customer retention rate
 - Generate business recommendations
 - Create professional visualizations and dashboards
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3. Dataset Description

Customer Dataset

The customer dataset contains customer-related information such as:

- Customer ID
- Contract type
- Monthly charges
- Total charges

- Payment method
- Churn information

Sales Dataset

The sales dataset contains transaction-related information such as:

- Date
- Product name
- Quantity sold
- Product price
- Region
- Total sales amount

4. Tools and Technologies Used

Tool/Technology Purpose

Python	Programming Language
Pandas	Data Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization

5. Data Preprocessing

The following preprocessing steps were performed:

- Loaded datasets using Pandas
- Checked dataset structure
- Identified missing values
- Converted date columns into datetime format
- Created calculated columns
- Renamed columns for merging
- Cleaned and formatted data

6. Statistical Analysis

Descriptive Statistics

The project used descriptive statistical methods to summarize data:

Statistical Method Purpose

Sum	Total revenue calculation
Mean	Average sales calculation
Maximum	Identify top customers/products
Count	Count total transactions
Correlation	Analyze product relationships

7. Aggregation Analysis

The project implemented multiple aggregations using Pandas.

Aggregation 1: Total Revenue by Region

```
sales_df.groupby('Region')['Total_Sales'].sum()
```

Aggregation 2: Average Quantity by Product

```
sales_df.groupby('Product')['Quantity'].mean()
```

Aggregation 3: Maximum Customer Revenue

```
sales_df.groupby('Customer_ID')['Total_Sales'].max()
```

Aggregation 4: Order Count

```
sales_df.groupby('Customer_ID')['Product'].count()
```

8. Data Merging / Joining

The customer and sales datasets were merged using Pandas merge() function.

```
merged_df = pd.merge(  
    sales_df,  
    customer_df,  
    on='Customer_ID',  
    how='left'  
)
```

This allowed combined customer and transaction analysis.

9. Pivot Table Analysis

Pivot tables were used to summarize sales performance.

```
pivot_table = pd.pivot_table(  
    sales_df,  
    values='Total_Sales',  
    index='Region',  
    columns='Product',  
    aggfunc='sum'  
)
```

Purpose:

- Compare products across regions
 - Identify top-performing product categories
 - Simplify sales summarization
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10. Customer Analysis

The following customer analyses were performed:

- Top customer identification
- Customer Lifetime Value (CLV)
- Customer retention analysis
- Regional customer distribution

Key Findings

- Some customers contribute significantly to total revenue
 - Retained customers generate higher long-term revenue
 - Customer behavior differs across regions
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11. Sales Pattern Analysis

Sales trends were analyzed using monthly and regional grouping.

Analysis Performed

- Monthly sales trend analysis
- Product sales analysis
- Regional performance analysis

- Seasonal trend analysis

Key Findings

- Sales fluctuate seasonally
 - Certain products dominate overall revenue
 - Some regions outperform others
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12. Dashboard and Visualization

Professional visualizations were created using Matplotlib and Seaborn.

Visualizations Included

- Line Chart → Monthly Sales Trend
 - Bar Chart → Best Selling Products
 - Pie Chart → Regional Sales Distribution
 - Heatmap → Region vs Product Sales
 - Countplot → Customer Churn Analysis
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13. Technical Requirements Fulfilled

Requirement	Status
Pandas Data Manipulation	Completed
Multiple Aggregations	Completed
Data Merging/Joining	Completed
Pivot Tables	Completed
Professional Visualizations	Completed

14. Business Insights

The project generated the following insights:

1. Top customers contribute a major portion of revenue.
2. Product demand changes across seasons.
3. Regional sales performance varies significantly.
4. Customer retention improves long-term profitability.
5. Product bundling can improve cross-selling opportunities.

15. Recommendations

Based on the analysis, the following recommendations were made:

- Introduce loyalty reward programs
- Improve customer retention strategies
- Create bundled product offers
- Increase marketing in high-performing regions
- Use seasonal trends for inventory planning

16. Conclusion

The Customer Sales Analysis Project successfully demonstrates the use of Python and Pandas for statistical analysis, business intelligence, and dashboard creation.

The project fulfills all technical requirements including:

- Data manipulation
- Aggregation analysis
- Data merging
- Pivot table analysis
- Dashboard visualization

The generated insights can help businesses improve customer retention, optimize marketing strategies, and increase sales revenue.